Impacts of Decoder Latency on Utility-Scale Quantum Computer Architectures

Abdullah Khalid, Allyson Silva, Gebremedhin A. Dagnew, Tom Dvir, Oded Wertheim, Motty Gruda, Xiangzhou Kong, Mia Kramer, Zak Webb, Artur Scherer, Masoud Mohseni, Yonatan Cohen, and Pooya Ronagh

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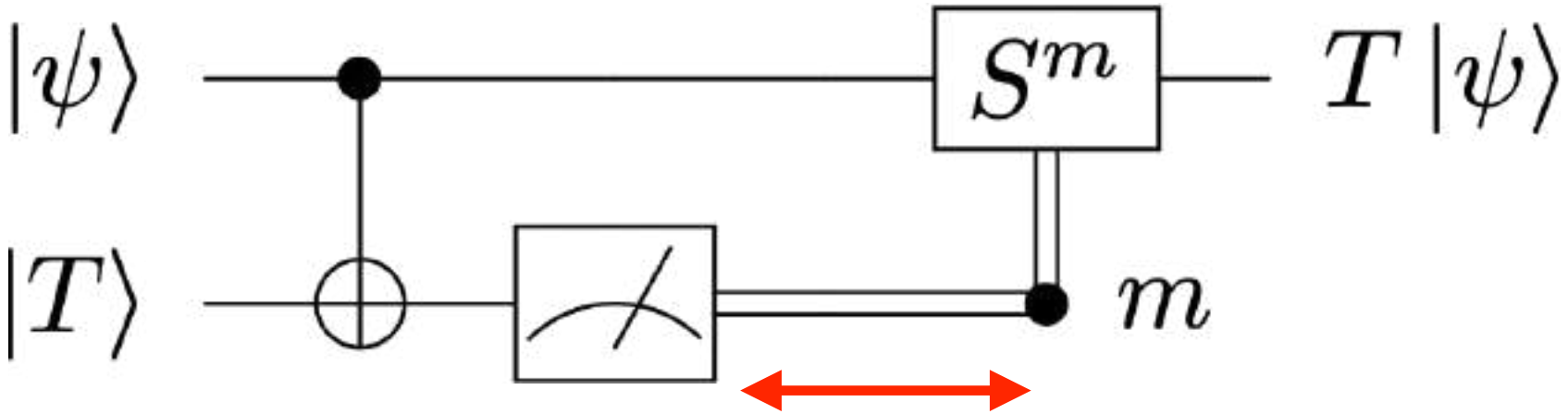
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Decoder Latency / Reaction Time

Sum of all time spent for classical transmission and processing



What is the impact of finite reaction time on the performance of a quantum computer?

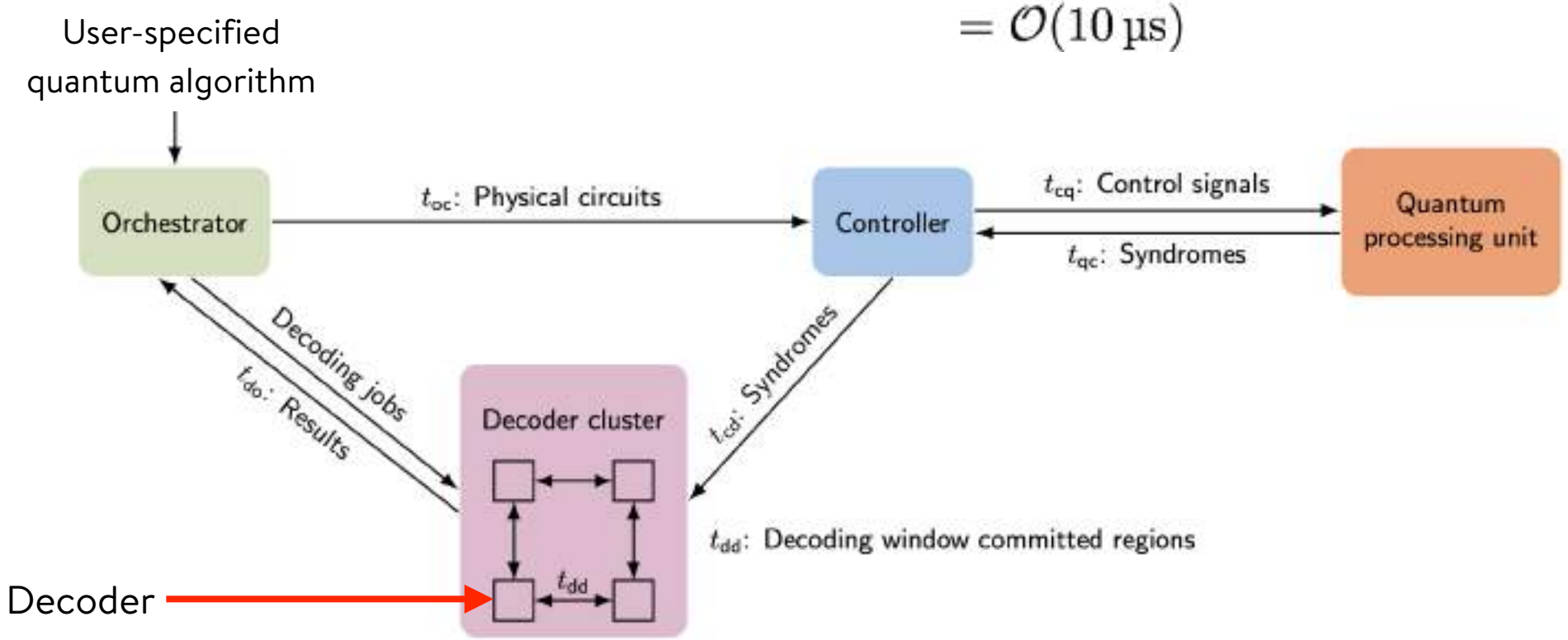
Gidney and Fowler, arXiv:1905.08916 (2019).

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Importance of Communication Latency

A system architecture for a quantum computer

$$\tau_{\text{com}} \approx t_{\text{qc}} + t_{\text{cd}} + t_{\text{dd}} + t_{\text{do}} + t_{\text{oc}} + t_{\text{cq}} = \mathcal{O}(10\,\mu\text{s})$$

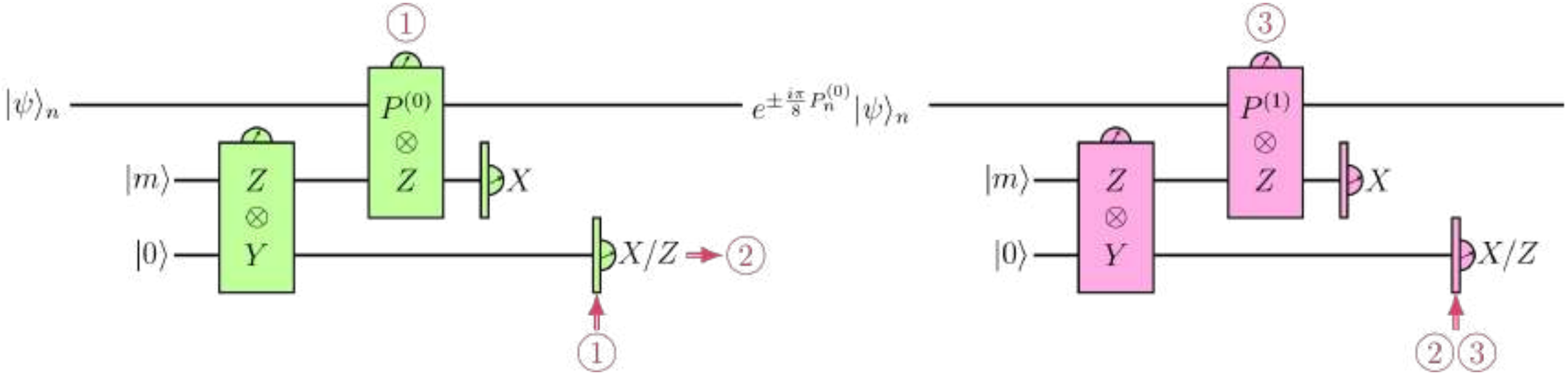


reaction time = decoder latency + communication latency

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Quantum Computation in Litinski's Architecture

A sequence of $\pi/8$ rotation gadgets



The choice of basis is determined as follows:

1. Joint qubit measurement (reaction 1)
2. The previous gadget's ancilla measurements (reaction 2)

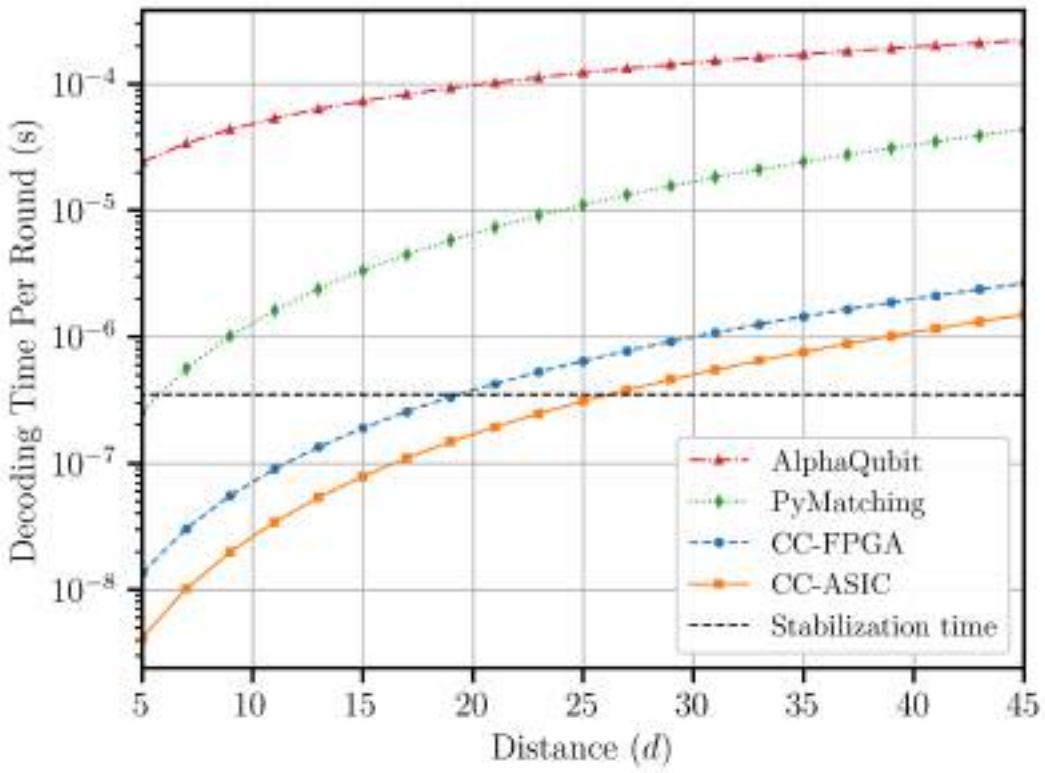
Litinski, Quantum **3**, 128 (2019).

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State-of-the-Art Decoding Times

Too large to perform FTQC

Decoding a d^2 Patch



Decoder	Decoding time per round $\tau_d(N) = \alpha N^\beta$
Collision Cluster on ASIC (CC-ASIC) [1]	$5.53 \times 10^{-11} N^{1.34}$
Collision Cluster on FPGA (CC-FPGA) [1]	$2.85 \times 10^{-10} N^{1.2}$
AlphaQubit [2]	$4.8 \times 10^{-6} N^{0.503}$
PyMatching [2]	$5.91 \times 10^{-9} N^{1.17}$

- State-of-the-art decoders are far too slow.
- Long memories or large lattice surgeries can have decoding times spanning dozens of logical cycles.

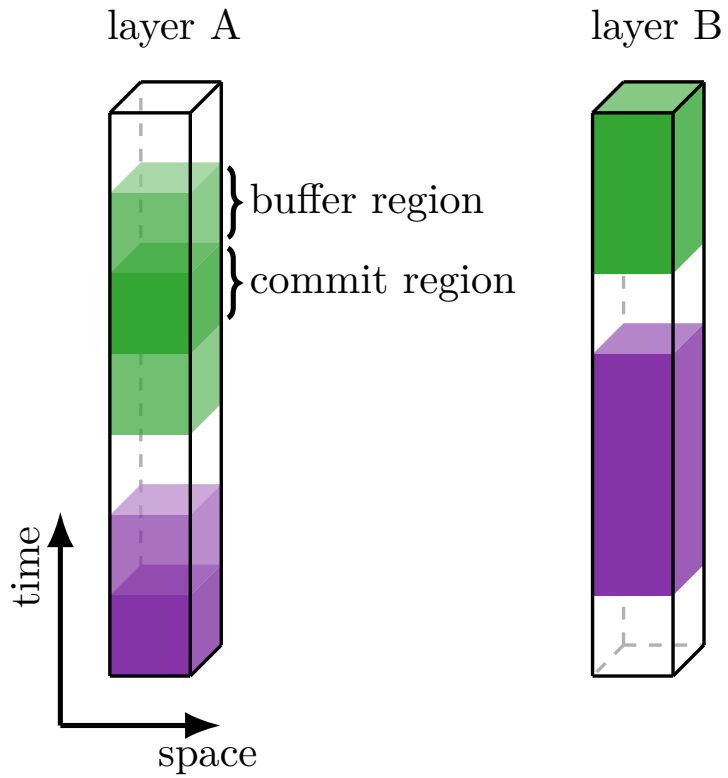
[1] Barber et al., Nature Electronics **8**, 84 (2025).
[2] Bausch et al., Nature **635**, 834 (2024).

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Decoder Latency with Parallelization Algorithms

Memory

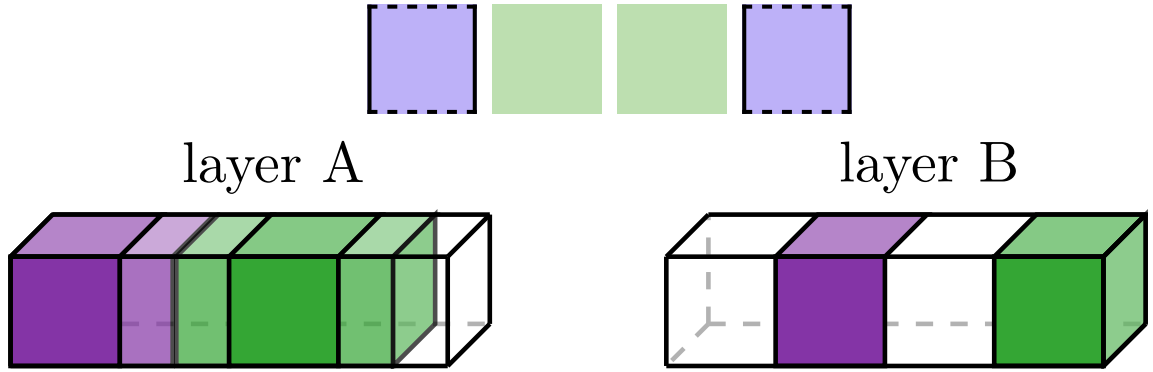
$$\gamma_{\text{mem}} = 6d\tau_d(d^2) + t_{\text{com}}$$



Decoding windows of temporal size: $d + d + d$

Lattice surgery

$$\gamma_{\text{LS}} = 3d(4^\beta + 3^\beta + 1)\tau_d(d^2) + t_{\text{com}}$$



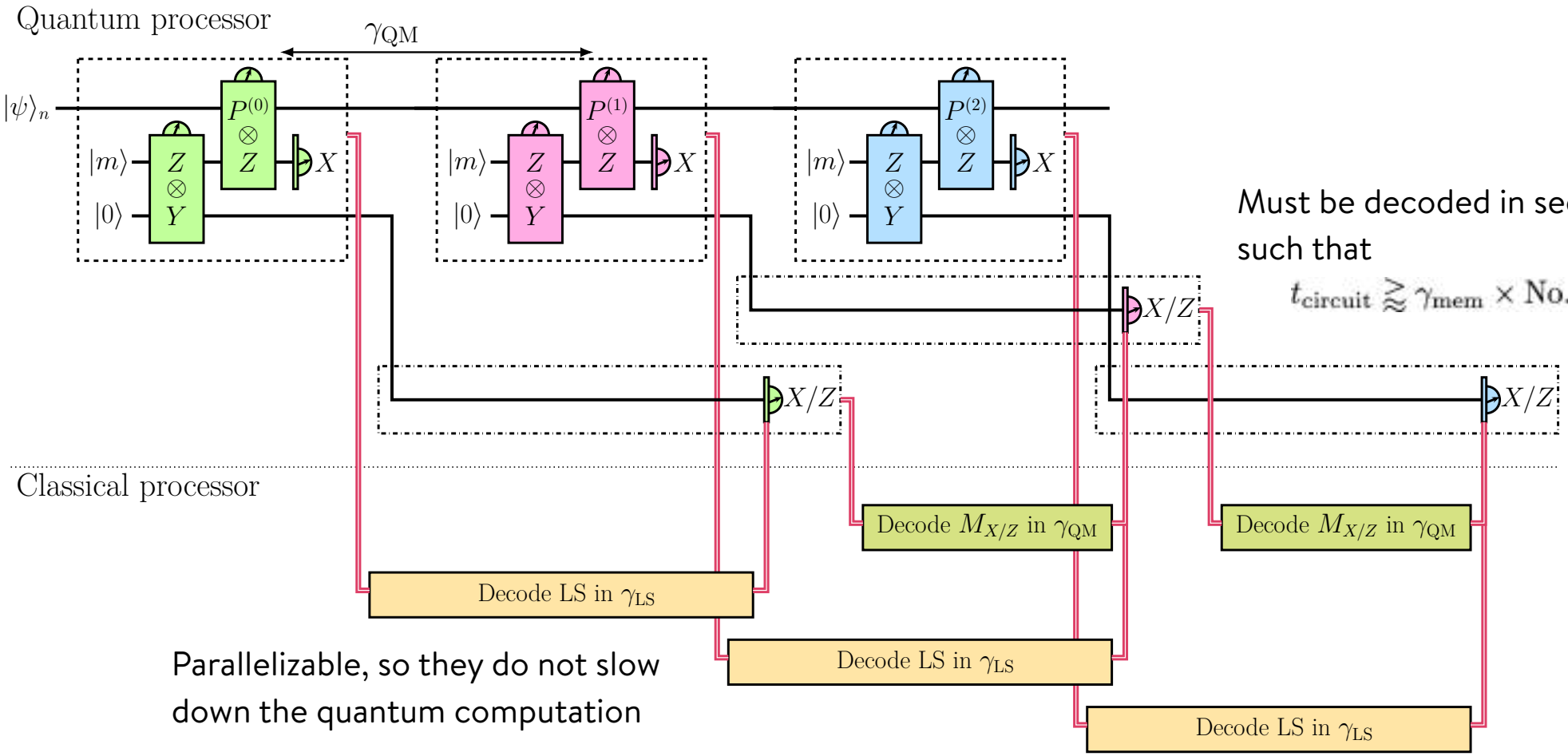
Decoding windows of spatial size: $d/2 + d + d/2$

With parallelization, decoder latency is a function of only the distance, and not the number of patches or rounds.

Lin et al., Quantum Sci. Technol. **10**, 035007 (2025).
Skoric et al., Nat Commun **14**, 7040 (2023).

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Impact of Reaction Times on Computation



Must be decoded in sequence such that

$$t_{\mathrm{circuit}} \gtrapprox \gamma_{\mathrm{mem}} \times \text{No. } T \text{ gates}$$

Parallelizable, so they do not slow down the quantum computation

$$\mathrm{LER}_{\mathrm{gadget}} \approx \mathrm{LER}_{\mathrm{LS}} + \underbrace{[n\gamma_{\mathrm{mem}} + (\gamma_{\mathrm{LS}} + 1)]\mathrm{LER}_{\mathrm{mem}}}$$

Additional error due to reaction times

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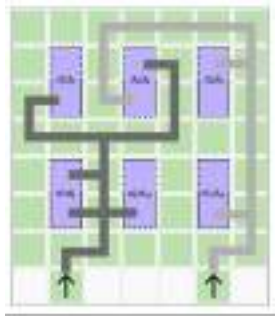
TopQAD™ Software Suite

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Quantum algorithm

Compiler

Operation schedule in processing unit

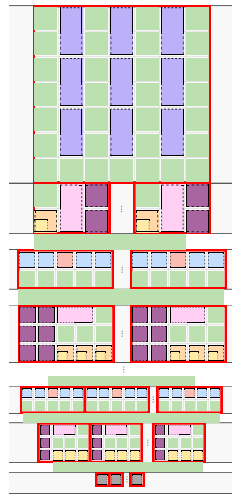


Decoder characteristics

Assembler

1. Number, location, and code distance of each unit in the logical microarchitecture

2. Full schedule of operations in the QPU



Noise model of QPU

Noise Profiler

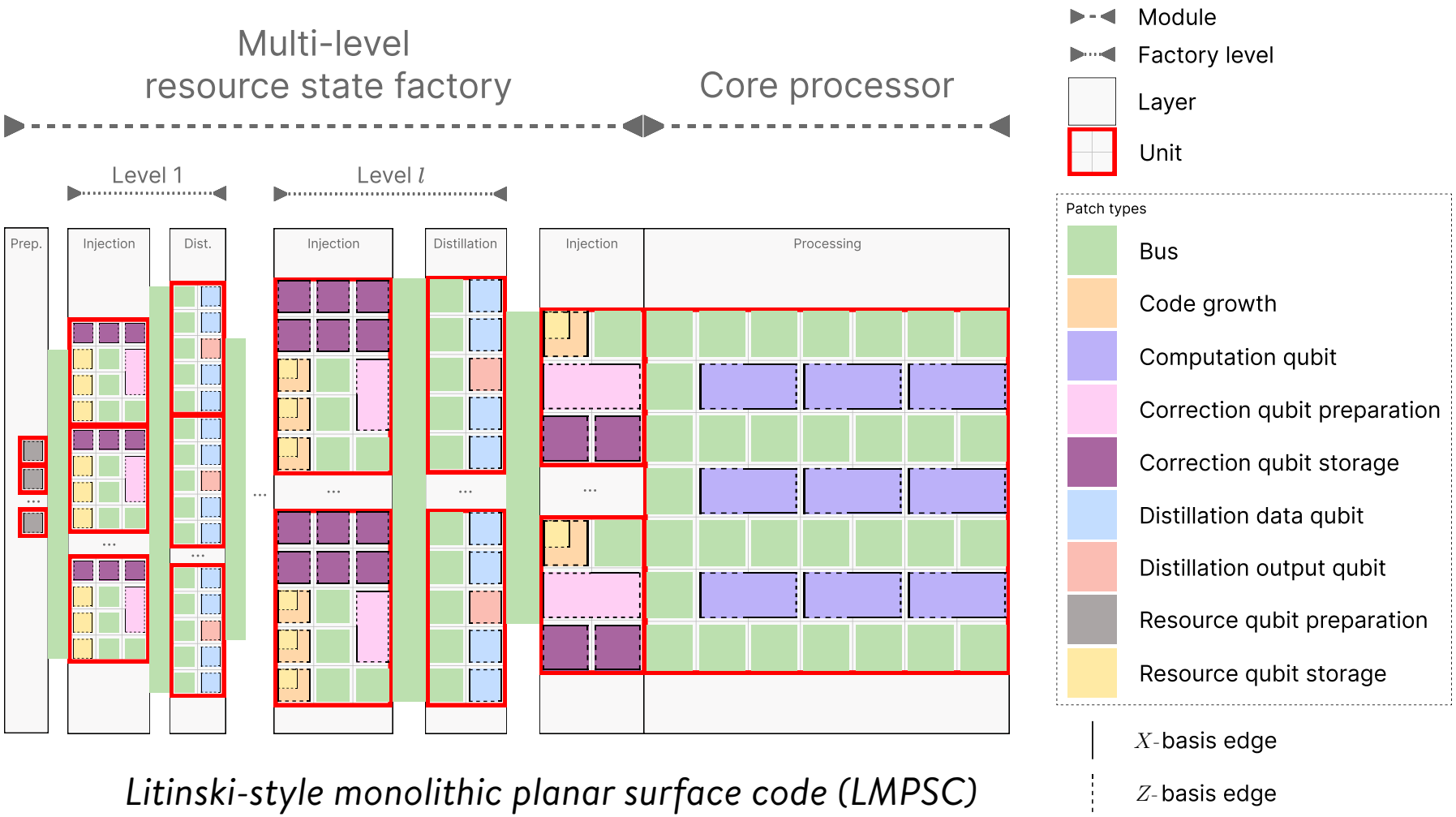
Performance characteristics of all FTQC protocols

1QB Information Technologies (1QBit), TopQAD: Topological Quantum Architecture Design [Software Documentation], (2024). https://topqad.1qbit.com/, https://doi.org/10.70781/YPWW8761

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Logical Microarchitecture

A specific arrangement of logical modules and submodules in the QPU that enable the execution of a quantum program.



Litinski-style monolithic planar surface code (LMPSC) logical microarchitecture family

Litinski, Quantum **3**, 128 (2019).
Khalid et al., arXiv:2511.10633.

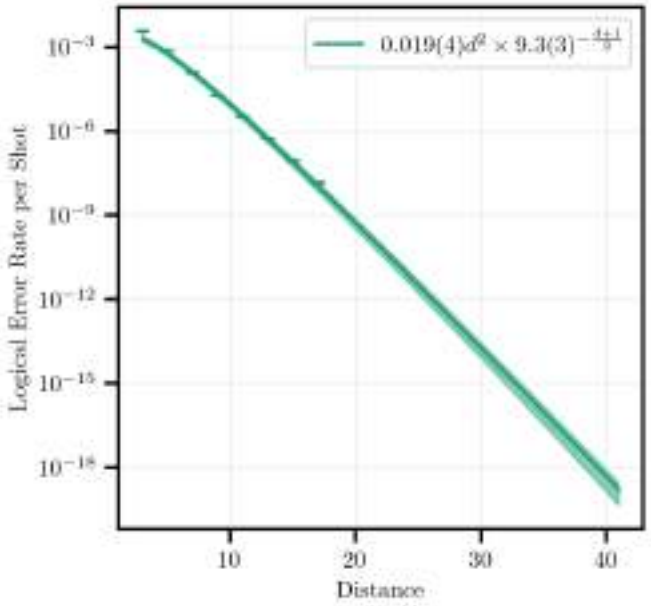
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The QPU Noise Profile

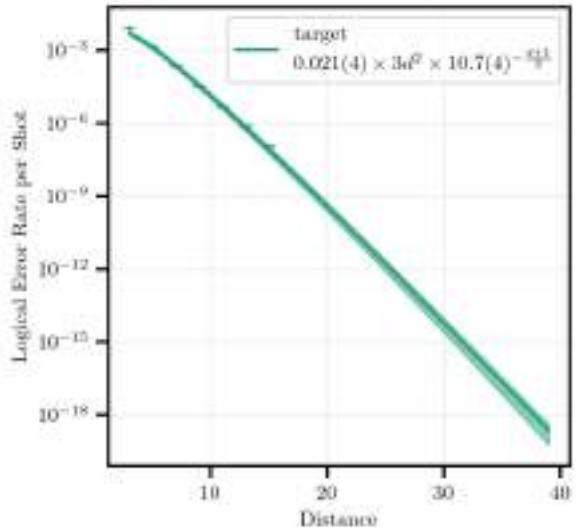
Target hardware parameter set [1] for superconducting qubits

Hardware parameter	Value
One-qubit gate error p_{g1}	0.0002
Two-qubit gate error p_{g2}	0.0005
State preparation error p_p	0.01
Reset error p_r	0.005
Measurement error p_m	0.005
T_1, T_2 times	200 µs
One-qubit gate time t	25 ns
Two-qubit gate time t	25 ns
State preparation time t	1 µs
Reset time t	100 ns
Measurement time t	100 ns

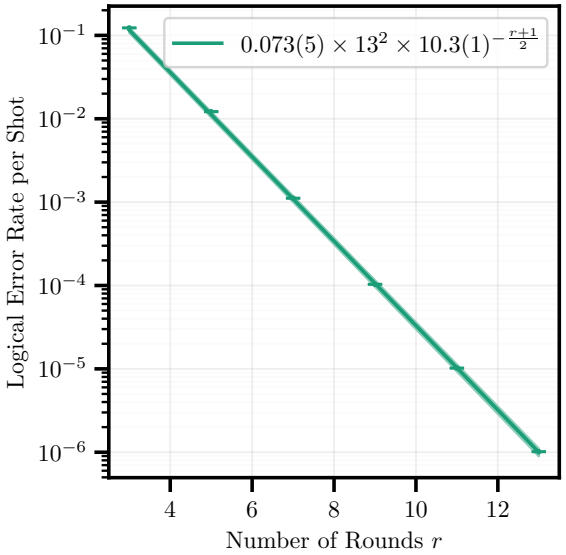
Memory Distance Scaling



LS Distance Scaling



LS Rounds Scaling



Magic state cultivation: LER ~ 4.73×10^{-5}

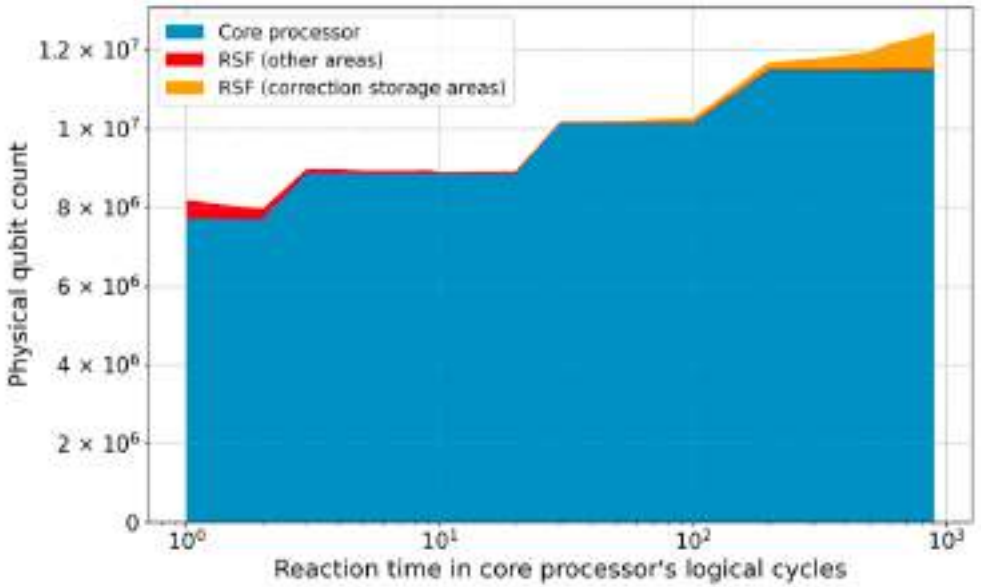
Discard rate ~ 0.41

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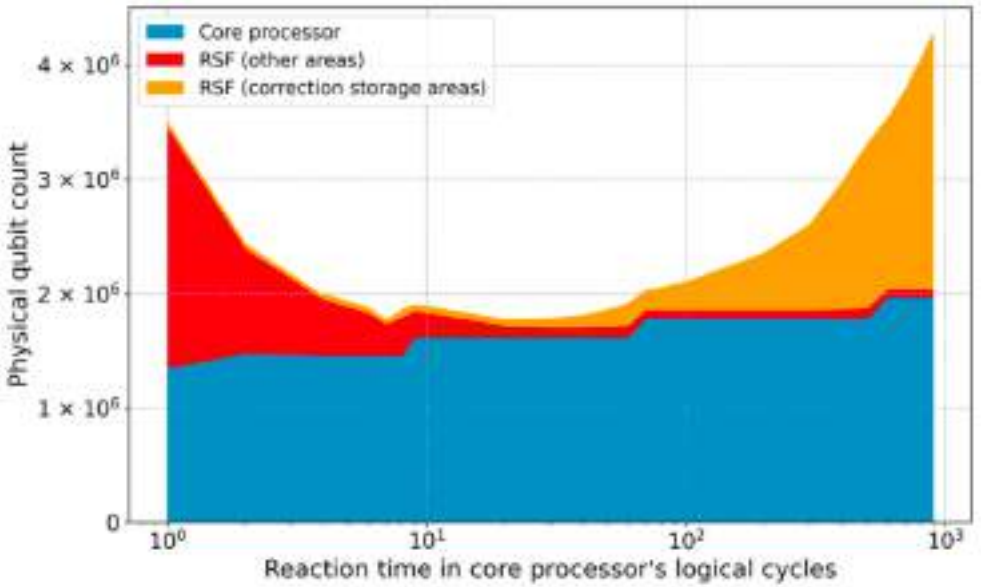
[1] Mohseni et al., arXiv:2411.10406 (2026).

Quantum Resource Estimation with Varying Reaction Time

Time-optimal solutions



2D Fermi–Hubbard model: 32×32 lattice
2562 logical qubits and 4×10^6 T gates



NMR spectral prediction for α-conotoxin
241 logical qubits and 5.11×10^{11} T gates

- reaction time $\div 2 \implies$ number of distillation units $\times 2$
  $\implies$ correction qubit storage $\div 2$
- reaction time $\div \Lambda \implies$ distance $- 2$

[1] Mohseni et al., arXiv:2411.10406 (2026).

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Time Required to Execute Different Algorithms

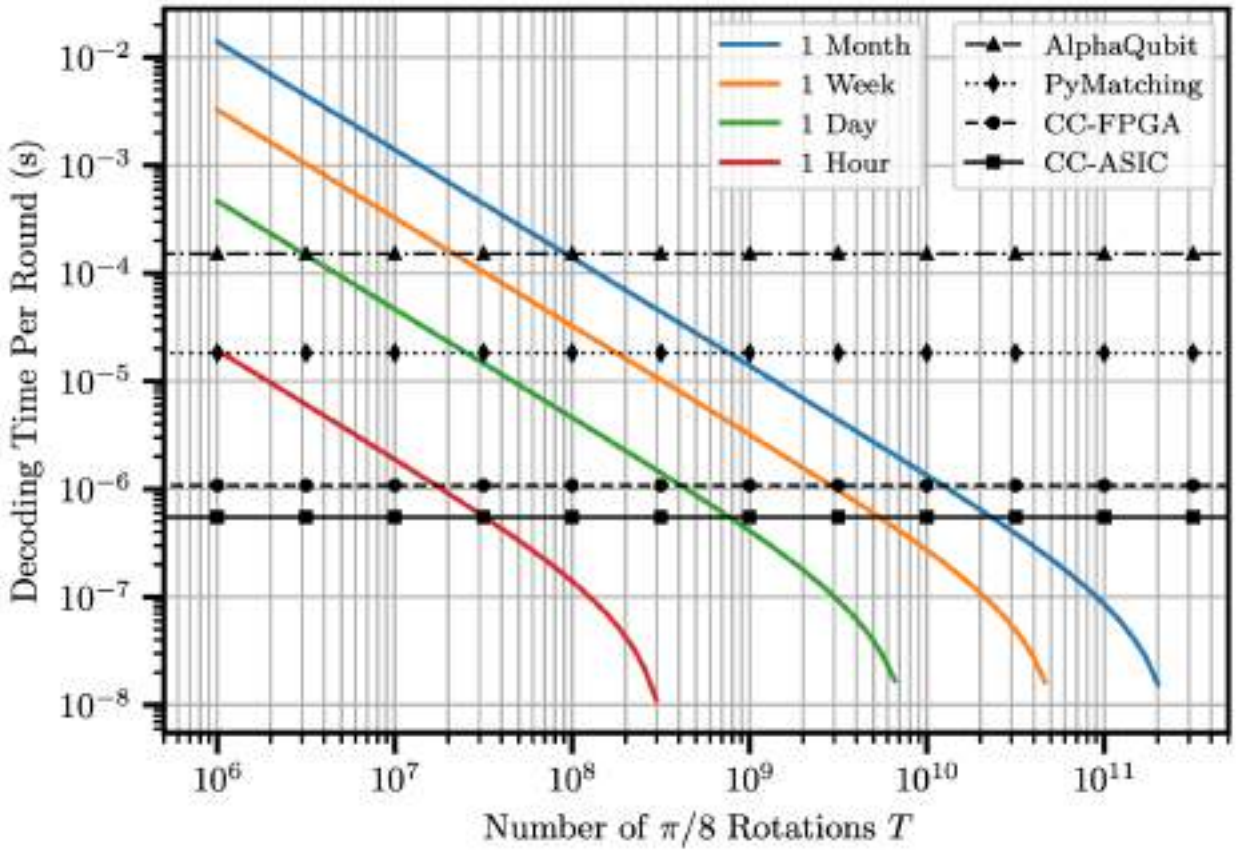
Given a fixed decoder

$$t_{\text{circuit}} = \gamma_{\text{mem}} T$$

$$\implies \tau_d(d^2) = \frac{t_{\text{circuit}}/T - t_{\text{com}}}{6d}$$

The fastest SOTA decoder can execute only 3×10^7 T gate circuits in 1 hour.

Decoding Times Incorporating Parallelization



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Conclusions

- The latency of decoding quantum memories is the main classical bottleneck in quantum computation.
- Reaction time can have a significant impact on the physical qubit count because of an increase in logical error rates.
- Decoders must be improved significantly to be able to execute quantum algorithms at utility scale.

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Thank You.

Khalid et al., Impacts of Decoder Latency on Utility-Scale Quantum Computer Architectures, arXiv:2511.10633 (2025).

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Noise Model

Operation $\mathcal{O}$	Noisy operation
One-qubit gate	$\mathcal{D}_1\left(\frac{3}{2}p_{g1}\right) \circ \mathcal{O}$
Two-qubit gate	$\mathcal{D}_2\left(\frac{5}{4}p_{g2}\right) \circ \mathcal{O}$
State preparation in $\|0\rangle$	$\mathcal{F}_X(p_p) \circ \mathcal{O}$
Reset to $\|0\rangle$	$\mathcal{F}_X(p_r) \circ \mathcal{O}$
Measurement in Z basis	$\mathcal{O} \circ \mathcal{F}_X(p_m)$
Idle on qubit for time t	$\mathcal{D}_1\left(\frac{3}{4}(1-e^{-t/T_1})\right)$

Hardware parameter	Value
One-qubit gate error p_{g1}	0.0002
Two-qubit gate error p_{g2}	0.0005
State preparation error p_p	0.01
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State preparation time t	1 µs
Reset time t	100 ns
Measurement time t	100 ns

Name	Operation	Constraint
One-qubit depolarizing $\mathcal{D}_1(p)$	$(1-p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z)$	$0 \leq p \leq 0.5$
Two-qubit depolarizing $\mathcal{D}_2(p)$	$(1-p)\rho + \frac{p}{15}\sum_{P\in\{I,X,Y,Z\}^{\otimes 2}\setminus II} P\rho P$	$0 \leq p \leq 0.75$
One-qubit bit-flip channel $\mathcal{F}_X(p)$	$(1-p)\rho + pX\rho X$	$0 \leq p \leq 1$

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